Full Title of trial:

Improving access to Package of Essential services for severe Non-communicable Diseases in Tanzania- PEN PLUS PROJECT

Acronym:
TANZANIA PEN PLUS PROJECT

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Authorised by:

Name	Dr. Mary Theodory Mayige
Role	Chief Investigator
Signatures	
Dates	15th /October/2022

GENERAL INFORMATION

This document describes procedures for evaluation of an implementation of PEN PLUS package of essential NCDs interventions at Kondoa and Karatu districts in Tanzania. The protocol should not be used as an aide-memoire or guide for the management of patients; every care was taken in its drafting, but corrections or amendments may be necessary for improvements as per our Tanzanian settings. The document will be circulated to the team of investigators. Any challenge relating to this trial should be referred to the relevant Chief Investigator.

Compliance: The trial will be conducted in compliance with international Good Practice Guidelines. The trial will adhere to the National Health Research Ethics Committee (NatHREC) in Tanzania.

List of the project investigators and technical working group	
Name	**Organization**
Dr Mary Mayige	NIMR
Prof Julie Makani	MUHAS
Dr Amani Mori	MUHAS
Mr.Gibson Kagaruki	NIMR
Dr Peter Kisenge	JKCI
Prof. Gene Bukhman	Brigham and Women's Hospital
Dr. Alma Adler	Brigham and Women's Hospital
Dr. Emily Wroe	Brigham and Women's Hospital
Dr Willfredius Rutahoile	Benjamin Mkapa Hospital
Dr Bruno Sunguya	MUHAS
Dr Reuben Mutagaywa	MUHAS/JKCI
Dr Pilly Chillo	MUHAS
Dr Edna Majaliwa	MNH
Dr Aidan Banduka	MNH
Prof Kajiru Kilonzo	KCMC
Dr Sarah Urasa	KCMC
Dr Deogratias Soka	TANSCDF
Dr Michelle D'Alessandro	DOCTORS WITH AFRICA CUAMM
Dr Esther Mtumbuka	CHAI
Prof Kaushik Ramaiya	TANCDA/ TDA
Dr David Raymond Mwenesano	NHIF

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List of abbreviation

CHMT	Council Health Management Team
CHW	Community healthcare workers
DALYs	Disability-adjusted life years
HIMS	Health Information Management System
NCDIs	Noncommunicable diseases and injuries
NCD	Noncommunicable diseases
PEN	Package of Essential Noncommunicable Disease
RHMT	Regional Health Management Team
RHD	Rheumatic heart disease
SCD	Sickle cell disease
TDA	Tanzania Diabetes Association
WHO	World Health Organization

Abstract

The burden of noncommunicable diseases and injuries (NCDIs) is increasing globally and Tanzania being inclusive. According to WHO, In Tanzania, non-communicable diseases (NCDs) are estimated to contribute up to 33% of all deaths and are projected to surpass the burden of deaths from communicable diseases in the next few years. Recently, the National NCDI Commission has found that NCDIs are a major cause of disease in Tanzania, contributing up to 41% of all DALYs, and that the burden of NCDIs has doubled in the past two and half decades. Both Communicable and NCDIs possess a threat to an already disrupted health care delivery in Tanzania. Increase in demand and high cost in accessing high quality NCDs care remain to be a challenge. For instance, the high cost of care, particularly at the secondary and tertiary levels, leaves many who are in need to fail accessing appropriate management. Because of this, there is a need to improve quality and improve care based on standardized protocols.

Therefore, a team of investigators decided to come up with WHO Package of Essential Noncommunicable Disease Interventions (WHO PEN) and PEN Plus model of NCDIs care to test, generate evidence and its practicality to our environment. The model provides an opportunity to improve the quality of care to first referral level facilities where the majority of the community members do attend for health care. It provides more support to community and primary care levels, focusing on prevention, early diagnosis, treatment, and reducing complications, which tend to be costly to individuals, and the health care system.

PEN and PEN Plus model will be implemented at Kondoa and Karatu districts in Tanzania, the two districts in different zones to establish capacity to manage severe NCDs. The proposed sites have been selected based on the burden of diseases associated with selected NCDs (Type 1 diabetes, SCD, and RHD) and the existing capacity of the zonal referral hospital to support the study implementation. This will be a pre- and post-intervention trial in which both quantitative and qualitative methods will be used to collect data from different study groups. We anticipate to recruit 528 households for the community surveys using multistage cluster sampling strategy. This study will be conducted in 11 wards, 22 villages/streets and 528 households. In each selected household, an interviewer will list all eligible households' members i.e. those aged 25-64 years and using Kish method, one eligible member will be selected, Kish method help to balance the age and sex of respondents for the entire study. On the other hand, all patients with Type 1 diabetes, SCD and RHD at the facility will be enrolled, and will have 24 key informant interviews and 8 focus group discussions. The project will be done using the well-known PEN PLUS toolkit, with changes made at all levels to meet local needs while staying true to the key principles. Implementation of this project will address the gaps and generate evidence from decentralized NCD services for the severe chronic condition, and also progressively to meet the need for early diagnosis and management of NCDIs by creating community awareness and improving detection capacity at lower-health facilities in Tanzania.

Background

The burden of noncommunicable diseases and injuries (NCDIs) is increasing globally. In Tanzania, non-communicable diseases accounted for 33% of all deaths (WHO 2018) and are projected to surpass the burden of deaths from communicable diseases in a few years. The National NCDI Commission reviewed and analyzed national representative survey data and found that NCDIs are a major cause of disease in Tanzania, accounting for 41% of all DALYs, which include disability and premature death, and that the burden of NCDIs has doubled in the past 25 years (NIMR 2020a).

Because of this, the National Institute for Medical Research (NIMR) has made NCD research a key part of its institutional strategic plan for 2019–2024 (NIMR 2020b) and the National Health Research Agenda (NHRA 2021 Draft).

Even though it is clear that NCDI services are part of the essential services package, the number of services in the country does not match the number of people who need them, and there is a lot of difference between rural and urban areas. There is evidence that availability and readiness are lower in rural areas than in cities and areas close to cities. The report also highlighted barriers to accessing treatment for certain conditions, such as hypertension and diabetes, especially in rural and lower socioeconomic groups (NIMR 2020a). Significant improvements in diabetes were observed where specific interventions, such as the National Diabetes Program, were implemented. The disparity was noted in both the availability and access to care as well as the availability of trained health care workers. The National Diabetes Program was put in place with help from the Tanzania Diabetes Association (TDA). Since 2003, the TDA has used programs related to diabetes and noncommunicable diseases (NCD) and helped set up services for diabetes and NCD in all national, zonal, regional, and district hospitals. TDA has also introduced a program for children with type 1 diabetes, and currently more than 2,000 children are accessing treatment at Zonal, Regional, and District Hospitals.

The high cost of care, especially at the secondary and tertiary levels, is one of the main problems with providing services for NCDs in the country. Because of this, there is a need to improve quality and improve care based on standardized protocols. WHO PEN and PEN Plus provide an opportunity to shift the paradigm of care to first referral level facilities and provide more support to community and primary care levels, focusing on prevention, early diagnosis, treatment, and reducing complications, which tend to be costly to the patient, family, and health care system (Bukhman 2020). PEN is an abbreviation for a Package of Essential Noncommunicable Disease Interventions at the primary health care facilities, while

PEN Plus, is a buildup of PEN package to manage severe NCDs at District Hospitals which are regarded as the first level referral facilities from the primary health care facilities (Bukhman 2020). For more details of the PEN and PEN Plus model, refer to the figure 1 below.

In order to ensure equitable access to NCD services, more investment is needed to improve the coverage and quality of NCD care services such as those for diabetes, cardiovascular diseases, cancers, chronic respiratory diseases, mental health, and severe NCDs that mostly affect the poor, such as rheumatic heart diseases (RHD) and heart failure, type 1 diabetes, and sickle cell disease (Bukhman 2020).

While hypertension has long been the focus of NCD programs in Tanzania, RHD is the third most common cause of heart failure after hypertensive heart disease and dilated cardiomyopathy. Patients with RHD present to health care facilities in late stages of the disease, when they already have complications such as stroke, heart failure, infective endocarditis, atrial fibrillation, and pulmonary hypertension. In 2010, 48% of patients who underwent open heart surgery were due to RHD, and the majority of these patients were young. The management of these patients at a late stage imposes a cost burden at the family level and in the health system and leads to poor treatment outcomes, including disability and deaths. In addition, patients often end up being referred to higher-level health facilities that are not easily accessible, especially to the poor (Carapetis 2005, Beaton 2012, Marijon 2007, Zuhlke 2015, Diao 2011, Makubi 2014).. Similar late presentations are seen across other disease entities such as diabetes, sickle cell disease, etc. (Piel 2013, Jasem 2019, Cannon 2017, Makani 2020, Madani 2011, Naghavi 2013, Makubi 2017, Makani 2015). An urgent need for early diagnosis and treatment through community awareness creation and improving detection capacity at lower-health facilities is therefore highly necessary (Bukhman 2020, Coates 2020, Eberly 2019).

Together with other institutions in the country and the Ministry of Health, the National Institute for Medical Research is in a good position to lead efforts to collect evidence to support the rollout of the PEN PLUS model of care for severe NCDs at first-level district hospitals and PEN services for other NCDs in primary care settings (health centers and dispensaries). This demonstration project will be embedded within the existing health system and infrastructure so that it can be sustained over time. We expect to generate robust data on the organization of care at first-level hospitals for underrepresented groups in cardiovascular care (RHD), type 1 diabetes, and sickle cell disease, for which the goal is improvement of care and potential national scale-up of the interventions.

Health services in Tanzania are organized into three levels, with dispensaries, health centers, and 158 district-level hospitals at the primary level, 27 regional hospitals at the secondary level, and tertiary-level health services provided by 12 zonal referral hospitals. national referral hospital. Although there are faith-based organizations and private health facilities in Tanzania, this proposal will focus on public health facilities, which provide healthcare to the majority of Tanzanians.

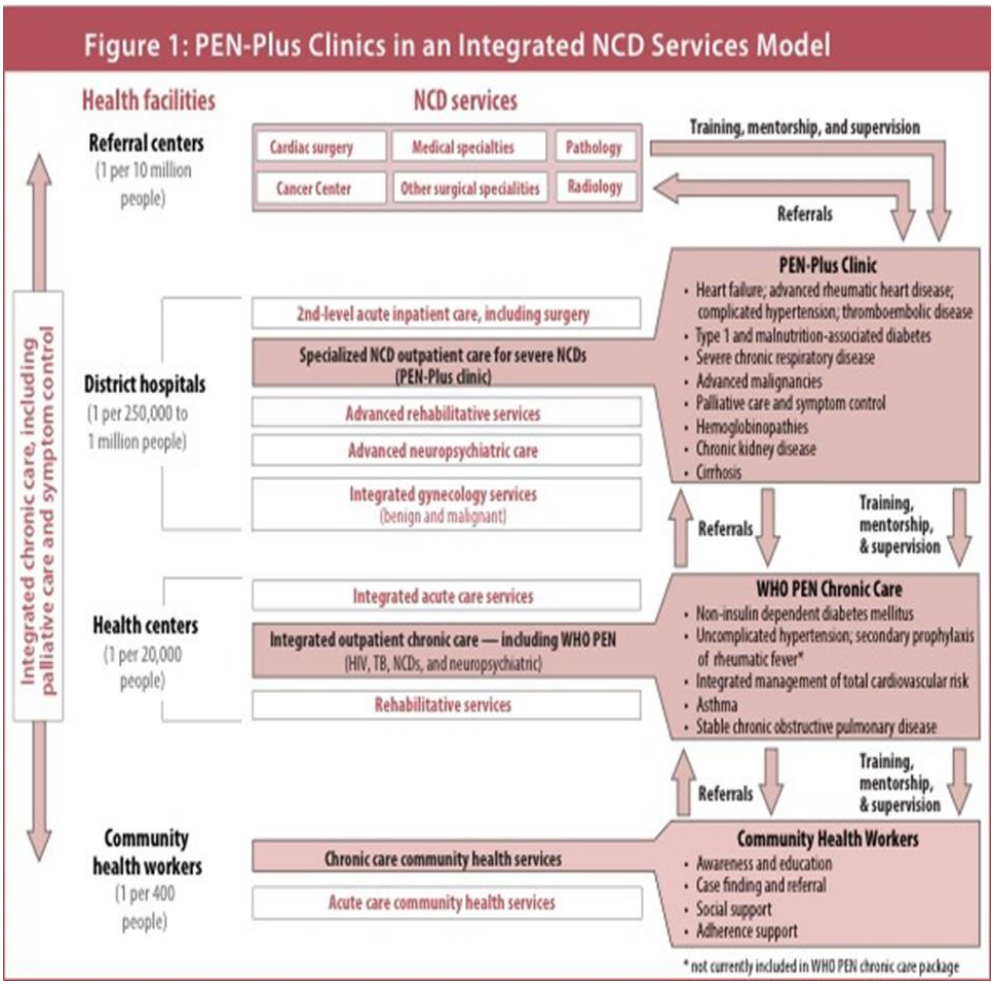


Figure 1: PEN PLUS Clinics, an integrated NCD service delivery model

Rationale

The Tanzania health system is overwhelmed by demand, particularly for acute and chronic illness due to infections such as diarrhoea, acute respiratory illness, malaria, HIV/AIDS, and TB. However, with the demographic transition, there is a growing burden of NCDs, which is posing a major challenge to a system that is already stretched. The majority of the health facilities are primary level facilities with limited capacity to provide care for non-communicable diseases, hence the significance of establishing PEN Plus clinics at first referral (district) level facilities that will provide support to the primary care facilities in managing stable patients and serve as a referral point for severe cases. Implementation of this project will address the gaps and generate evidence from decentralized NCD services for the severe chronic condition, and also progressively to meet the need for early diagnosis and treatment through community awareness creation and improving detection capacity at primary health care facilities.

Project Aim and Objectives

The main goal of the project is to find evidence and costed plans for expanding the WHO Package of Essential Noncommunicable Interventions (WHO PEN) for common NCDs and the PEN PLUS interventions for severe NCDs. This project will focus on three conditions at district (first-level) hospitals in Tanzania: type 1 diabetes, rheumatic heart disease, and sickle cell disease.

The objectives of this project are:

1. **To conduct baseline assessments facility capacity, care processes, and outcomes for severe non-communicable diseases at first-level referral hospitals in 2 districts.**
2. Conduct community surveys to establish the burden of NCDs.
3. To assess community and health care workers levels knowledge on NCDs at baseline and at end of project implementation.
4. To pilot , monitor and evaluate PEN-Plus services in 2 district hospitals **(both implementation and service outcomes)**
5. Establish disease registry for diabetes, hypertension (and heart failure) , SCD, rheumatic heart disease and assess patient outcomes in relation to baseline data and perceived benefits from the healthcare providers' perspective.
6. Establish cost effectiveness of PEN plus care /intervention package.

Methodology

Study sites

The project will be implemented in two districts in two different zones to establish capacity to manage severe NCDs. The proposed sites have been selected based on the burden of diseases associated with selected NCDs (Type 1 diabetes, SCD, and RHD) and the existing capacity of the zonal referral hospital to support the PEN Plus implementation. The model PEN PLUS hospitals will be established at district-level hospitals, which are the first referral facilities in the Tanzanian health system with a catchment population of about 200,000 to 300,000 people.

Table 1: Selected Project sites

ZONE	District	Justification
Central Zone	Kondoa District Hospital	High prevalence of RHD/ existing referral capacity at BMH
Northern Zone	Karatu District Hospital	High prevalence of RHD/ existing referral capacity at KCMC

Study design

This will be a pre- and post-intervention trial in which both quantitative and qualitative methods will be used to collect data from different study groups. This will be implemented from **December 2022 to November 2025**.

The project will be done using the well-known PEN PLUS toolkit, with changes made at all levels to meet local needs while staying true to the key principles. Therefore, in order to achieve the desired objectives, the following work plans will be implemented based on recommendations from the PEN PLUS implementation framework (Eberly 2019).

Study population and sample size calculation:

Community

The community is defined as the population within the chosen hospital catchment area.

Patients

Existing and newly diagnosed patients

Health facilities

District Hospitals from the two districts namely Kondoa and Karatu

Health care workers

Health Care workers in the care facility working on NCDs, including nurses, lab technicians, pharmacists, radiologists and CHMTs including district NCDs coordinators and RHMTs including regional NCD regional coordinator

NCD stakeholders

Professional Organizations e.g. Tanzania Cardiac Society, Endocrinology, Nephrology Society of Tanzania, Tanzania

Medical Association, Medical Council of Tanganyika (admin of CPD training)
Regional and District Health management Teams
Hospital Management Teams
Community Health Workers
Local Government Authorities
community and patients (via patient associations or advocacy groups)

Sample size calculations

Community Surveys

We propose to use the formula below to calculate the sample size of the PEN Plus baseline study.

$$n = D\frac{\left[Z_{1-\alpha}\sqrt{2P\,(1-P)} + Z_{1-\beta}\sqrt{P_1(1-P_1) + P_2(1-P_2)}\right]^2}{(P_2-P_1)^2}$$

We use a conservative value of 50% as the proposed baseline value of the indicator (P1), i.e., the percentage of community members with adequate NCD knowledge level; we also assumed an effect size of 10 percent for a P_2 of 62 percent and a power of 80 percent. The average of P_1 and P_2, i.e., P, is 56 percent. Other parameters for sample size calculation include the design effect (DE) of 2 (Julios 2020, Brown 2000, Fiorentino 2021). and a 10 percent attrition or non-response rate. Keeping in mind the above assumptions and the idea of proportional distribution, our actual sample size is 528. This sample size was distributed proportionally based on the population size in each target district, aged 25–64 years. In our case, in 2021, a total of 168,808 and 142,105 people aged 25–64 years were estimated from Kondoa and Karatu districts, respectively. Therefore, we will recruit 288 participants from Kondoa and 240 participants from Karatu.

Sampling procedure:

This survey will employ multistage cluster sampling that will involve three stages. The first stage will involve selection of wards, which is a primary sampling unit (PSU); a total of 11 wards (5 from Karatu and 6 from Kondoa) districts will be randomly selected. The second stage will involve randomly selecting two villages/streets in each selected ward, this stage is termed as a secondary sampling unit. The tertiary sampling unit will be the last stage of selection, in each selected village/street, 24 households will be selected randomly. To determine the starting household, the team will first go to some central location within the community e.g., village office, market, church, heath facility or junction between two roads and use the "spin bottle approach" to decide the direction. The nearest house will then be selected as the first house, then the next house will be subsequently selected after every three households. In summary this study will be conducted in 11 wards, 22 villages/streets and 528 households. In each selected household, an interviewer will list all eligible households members i.e. those aged 25-64 years and using Kish method, one eligible member will be selected, Kish method help to balance the age and sex of respondents for the entire study thus 528 individuals will be interviewed for the entire study.

Screening for Diabetes, RHD and SCD

Diabetes, RHD, and SCD will be checked for regularly at the facility and through outreach programs in schools and the community **during study period**. All people who agree to be screened will be screened, but the goal is to screen at least 25% of all people aged 6 and up in the target district councils. This target will be enough to display the full picture of the burden of the problem.

Inclusion: The inclusion criteria will include 1) Aged 6 years and above and 2)Consenting individuals

Exclusion criteria: The following will be our exclusion criteria: 1)Known SCD, 2)Diabetes, RHD patients, and 3) Non permanent resident

Diabetes, RHD, and SCD will be addressed in the target districts with all community members aged 6 and up. Estimates for

2021 indicate that 540,334(Karatu=250,576 and Kondoa=289,758) people were aged 6 years and above, therefore, we expect that this project will screen more that 135,000 people before the end of the project i.e.... 25% of the total population in the target councils

Key Informants: Data will be gathered from 24 key informants, as detailed below:
Community healthcare workers (CHW): four key informant interviews, two per district. In each district, an interviewer will work with the NCDs district coordinator to find two villages with competent CHWs. In each village, one CHW will be chosen at random to get information about NCD-related activities, problems they face when doing those activities, and ways to improve the way community-level NCDs activities are done.
Primary and secondary school health teachers: Eight key informant interviews will be conducted with health teachers. In each district, two health teachers from primary and secondary schools will be chosen at random and interviewed to find out what they know about noncommunicable diseases (NCDs), what NCD-related activities are done at their schools, what problems they face when doing these activities, how they have dealt with NCD patients at their schools, and what can be done to improve the quality of NCD services in primary and secondary schools.

District NCD Coordinators (DNCDCs): Two DNCDCs, one from each district, will be interviewed to find out what kinds and how many community NCDs activities they do, what problems they run into when doing these activities, and what can be done to improve community coverage of NCDs services.

District and Regional Hospital Pharmacists and Lab personnel: From the district and regional hospital levels, two pharmacists and two lab workers, one from each hospital level, collect information on the availability of NCDs equipment and supplies, the problems they have getting such equipment and supplies, and how to improve the supply and availability of NCDs equipment and supplies at different health facility levels.

District and regional HMIS/DHIS coordinators: Two HMIS coordinators, one from a district hospital and the other from a regional hospital, will be involved to collect information on the challenges they face in making sure the components of quality data (timeline, precision, completeness, reliability, consistency, standardization, and interoperability) are met, that there are enough resources to capture data, and that data collected at different levels is used.

National NCDs Coordinators: Two national-level NCD experts, one from the Ministry of Health and one from PORALG, will be interviewed about problems with the availability, accessibility, and cost of NCD prevention and treatment services, as well as ways to improve the quality of NCD services in Tanzania.

National Data Manager: For the MoH and PORALG, the study will collect data from the data managers and record information about the problems they face in ensuring the components of quality data (timeliness, accuracy, completeness, reliability, consistency, standardization, and interoperability), making sure there are enough resources to capture data, and making sure data collected at different levels of health facilities is used.

Focus Group Discussion: We plan to hold eight focus group discussions (FGDs) to learn about people's knowledge of NCDs, the availability, accessibility, and use of NCDs services, the problems people have getting to the services, and how to improve the availability, accessibility, and use of NCDs services. To gather the above information, the study will conduct two FGDs with primary school pupils, one per district, and two FGDs with secondary school students, one per district. Four FGDs will also be conducted with adult individuals, i.e., male and female, aged 18 and above, two FGD per district.

Allocation of study interventions
The implementation at the selected sites will take place in stages, beginning with one site and gradually expanding to

include a second site beginning in year two of implementation.

Project Scope

The selection of these conditions was guided by the severity of the conditions and epidemiological data. These conditions are a significant cause of morbidity and mortality, especially among children. The capacity strengthening will also cover other conditions such as type 2 diabetes and hypertension, but the provision of treatment support will be restricted to those with type 1 diabetes, sickle cell disease, and rheumatic heart disease due to limited resources.

Recruitment SCD, RHD, Diabetes Patients

Patient Recruitment

All new and old patients will be recruited to the established PEN Plus clinics, enrolled in the registry, and followed up for up to 2 to 3 years.

Recruitment of HCW

Health care workers involved in the care pathway of these patients will be identified, a baseline assessment will be done, and they will be trained on the PEN Plus care package.

Interventions Packages

Table 2: Disease Scope

Disease Group	Selected Condition
Cardiovascular	heart failure; advanced rheumatic heart disease; complicated hypertension; thromboembolic disease
Endocrine	type 1 and malnutrition-associated diabetes
Hematology	sickle cell disease

Interventions

Community

NCD Health Information: In order to deliver health education information to communities in the intervention areas, this study will use standard health information education and communication materials developed by MOH/TANCDA**.** The participants will be informed on the behavioral, familial, and metabolic risk factors for NCDs and prevention measures. These measures include the recommended amount of physical activity per week (MET/week), the amount of alcohol that each gender can drink in a week, and the amount of fruits and vegetables that each person should eat every day. Furthermore, the participants will be informed of the dangers of smoking and eating an unhealthy diet. The signs, symptoms, and complications of NCDs will also be explained to the community. There will be group meetings, text messages on their cell phones, and the handing out of brochures and banners as ways to spread important health messages.

Patients

The enrolled patients will receive routine care, diagnostics, routine follow-up tests, medications, and lifestyle counseling as per guidelines. Below is a table summarizing components of care for the conditions (standard of care only).

Table 3: Components of care

Disease system (or category)	Tests	Medications
Multi-specialty	radiography, blood pressure measuring devices, weight scale, measuring tape or ruler, hemoglobin, ESR, glucose, serum electrolytes, creatinine, PT/INR	benzathine penicillin, penicillin VK
Endocrine	blood glucose (including glucometers), monofilament, hemoglobin A1c	regular insulin, lente insulin, metformin, sulfonylureas

Cardiovascular, Renal, and Hepatology	electrocardiography, ultrasound equipment with cardiac and abdominal probes, liver function tests, hepatitis C and B testing	aspirin, furosemide, ACE-inhibitors, heart-failure beta-blockers, atenolol, spironolactone, thiazide diuretic, amlodipine, methyldopa, hydralazine, isosorbide dinitrate, warfarin, potassium, proton-pump inhibitors, lactulose and Losartan
Hematology	blood smear, full blood count, sickle scan	hydroxyurea, penicillin, folic acid

Health Care Workers

One important aspect of the implementation process is the training of health care providers and trainers for the successful rollout of WHO PEN/PEN PLUS implementation. The training will include both theoretical and practical lessons. Each of the district hospitals will receive the mentorship of a referral-specialized hospital. This will be done using different platforms and technologies to get the training materials to the people who need them. In this case, we will also use the existing platforms for continuing professional development (the MoHCDGEC has an e-learning platform on NCDs that can be used for this project), and a PEN PLUS Practical Training Center will be set up at one of the chosen intervention district hospitals.

Study the conceptual framework: Figure 2 shows a summary of the study's concept. The study is being done so that the effects of the PEN Plus care package can be evaluated and a nationwide scale-up can be planned. The **evidence will help to improve** care quality and patient outcomes by strengthening care inputs like procurement, supply chain management, training of health workers, availability of equipment and supplies, health information systems, and community and patient empowerment. This will help reach SDG 3 by reducing early deaths from non-communicable diseases.

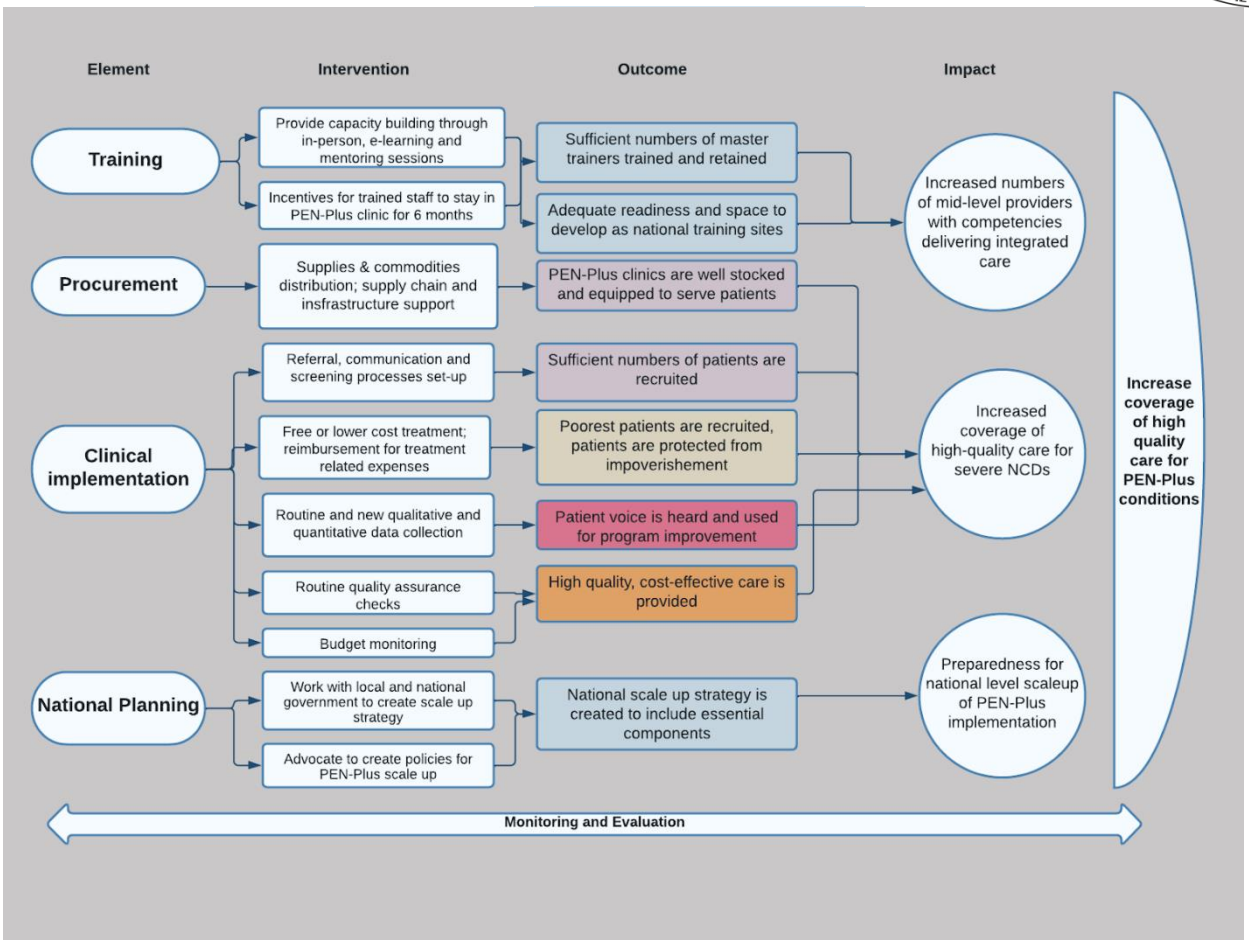


Figure 2: Conceptual frame work

Health facility and health care assessments will be collected using ODK software. Data will be collected electronically using the Open Data Kit. Before being synchronized in real time on the main server of the National Institute for Medical Research, data will be cleaned.
RedCap will be used to design databases for disease registries. Each person will be given a unique identifier that will protect their privacy.

Data analysis

Quantitative analysis: descriptive data analysis will be performed using Stata version 15.0 (StataCorp, 2017). Stata Statistical Software: Release 15. College Station, TX: StataCorp LLC. Both categorical and continuous variables will be compared before and after the intervention using Chi-Square tests for categorical variables and T-tests for continuous variables to see if there are any differences between the baseline and endline values. Explanatory variables will be categorized into two groups, namely random and fixed variables, to overcome dependence problems resulting from repeated measures. Generalized Linear Mixed Models (GLMM) will be used in the study to account for both fixed and random variables.

Qualitative Analysis: Depending on the size of the interview guide, we may use analytical software like NVivo, Atlas.ti, or Excel. We will use a combination of deductive and inductive approaches in an iterative manner. We will use a bottom-up approach to code the raw data, line by line. We will then develop themes and theoretical constructs by grouping the

base codes together as needed by the conceptual framework. Based on similar assessments, a first set of themes, categories, and codes could be chosen ahead of time. A coding template will be developed from a few of the early transcripts (including those from the pilot stage) and will be used to code later transcripts in an ongoing process as data is collected. Ongoing analysis will be distinguished by frequent returns to the original transcripts to ensure that text is coded within context. Following the coding process, themes and theoretical constructs will be finalized and compared across and within each sub-group to explore relationships between constructs and any similarities or disparities between groups. The data collector will work with the Community Health Workers (CHWs) and Village Executive Officers (VEOs) to find people to participate in FGDs. For school-aged children and students, the school heads will help with recruitment.

Ethical approval: This study will follow a set of ethical rules based on the revised Declaration of Helsinki from 1964. These rules include ethical approval and informed consent**.** The National Medical Research and Coordinating Committee will be asked to approve this study, and this protocol will also be added to the pan-African clinical trials registry. Every person who takes part in the study will be told its purpose and goals, as well as the risks and benefits of taking part. They will also be told that taking part in the study is completely voluntary, that it can be stopped at any time, and that not taking part in the study won't hurt them in any way. Each consenting participant will be asked to sign a consent/assent form. Interviews will take place in places where privacy and confidentiality will be protected. For example, the data collection questionnaires will be locked in a safe cabinet, and the electronic data file will be kept on a computer with a password. All study subjects diagnosed with any condition at screening will be informed and referred to a nearby health facility for further management.

Dissemination

Findings from routinely collected data will be shared with the local sites during M&E discussions every three months and formative assessments every six months. Progress will be mapped on a dashboard that sites can access at any time. We will send quarterly newsletters to all the sites highlighting statistics, progress, or successes, including characteristics and predictors of compliance. We keep in touch with national stakeholders, like the Ministry of Health, through regular feedback loops. Finally, we will disseminate our findings and best practice through publications in open access peer reviewed journals

STUDY IMPLEMENTATION

Baseline Assessment: assessment of burden and service coverage

Through the baseline assessment, the two sites for the pilot and demonstration sites will be selected and their current needs mapped. Standardized tools will be used to do a baseline assessment of the burden of certain noncommunicable diseases (NCDs) and the staff and equipment already in place in the health facilities and hospitals that are being targeted. This will help figure out what works and what doesn't, and give the baseline data needed to plan and evaluate project implementation in each district. This will help set the minimum requirements for putting WHO PEN/PEN Plus into place in Tanzania and will be the basis for future rollouts. It is proposed that the training site be at the implementation sites; if there is limited capacity, the training site will be established at the supervising referral facility.

Clinical Guideline Development

Existing NCD treatment guidelines will be reviewed to align with the recommendations of WHO PEN/PEN PLUS, with particular attention to the staffing qualifications, equipment, and supplies. This will help set the minimum requirements for implementing WHO PEN/PEN Plus in Tanzania and will be the basis for future rollouts. The implementation of this work will go through the engagement of professional bodies governing the various clinical specialties and will be validated through technical working groups with health workers of various cadres and specialties. The clinical protocol will also include community empowerment strategies (NCD advocacy, health education and early screening and referral; community referral pathways, and treatment support)

Equipment and Medications

PEN-Plus clinics use a basic set of equipment that is usually available at district hospitals and low-cost, generic medicines that are on the WHO's list of "essential medicines." Equipment includes radiologic modalities such as x-ray and ultrasound and laboratory devices for evaluating hemoglobin, creatinine, HIV, and blood glucose. In the table below, you can find more information about the PEN-Plus clinic's special tools, tests, and medicines. We will make sure the patients have access to the tests, medicines, and other supplies they need technology to get the training materials to the people who need them. In this case, we will use the existing platforms for continuing professional development (the MoHCDGEC has an e-learning platform on noncommunicable diseases that can be used for this project) to make sure the project will last. We will also set up a PEN PLUS Practical Training Center at one of the selected intervention district hospitals.

Staffing and Services

The clinic would have two or more clinicians who would split their time between clinical care, program monitoring and evaluation, and helping primary health centers provide services for people with chronic NCDs. This would make sure that the clinic was always open and that the services were of good quality.
In sub-Saharan Africa and other places with few resources, the best way to get long-term care is often through mid-level providers like clinical officers or nurses with advanced training. Specialized mid-level providers can give high-quality care for managing severe chronic diseases that need more attention to detail than common NCDs if they are empowered to do so. These mid-level providers will get regular supervision on-site from senior staff who live in the area and specialists who work at referral centers. The table below shows some specific clinical skills to address severe chronic NCDs.

Table 4 PEN-Plus Clinician Skills

Disease system (or category)	Specific Skills
Multi-specialty	proficiency with diagnostic and treatment protocols; palliative care and symptom control; counseling regarding home-based care; training, mentorship, and supervision of health center staff and community health workers
Endocrine	proficiency in insulin management; screening for complications of diabetes
Cardiovascular	capability to perform and interpret simplified echocardiography, abdominal ultrasound, and electrocardiography; ability to interpret basic chemistries; skill in management of heart failure medications and anticoagulants
SCD	capacity to conduct genetic counseling, diagnosis, and Management of common acute and chronic SCD complications, Screening for chronic SCD complications, including the conduct and interpretation of transcranial doppler (TCD),

Training and Supervision

In addition to classroom lessons, PEN-Plus providers will do a lot of supervised clinical work during their first training. Based on previous experience with this model, new providers can be trained in as little as three educational months once a clinic is up and running with a sufficient number of established patients. Educators will be trained in part by medical and pediatric specialists, with the intent that these mid-level providers will establish and independently manage PEN-Plus training sites. Following initial training, PEN-Plus providers will continue to receive continuous education and supportive supervision from specialists at higher-level hospitals as well as from more senior clinicians at their district or regional hospital.

Progressive Decentralization

With a high proportion of the population of Tanzania residing in rural settings, decentralization is necessary to provide

accessible care to remote communities. This work package will involve the development of a costed operational plan to enable decentralization of services, both through the expansion of PEN-Plus to additional district hospitals and through a plan to support the PEN-Plus hospitals to support the district-level facilities. By this means, support will be provided to lower-level health facilities, such as health centers, in order to establish basic NCD services at these lower levels, including supporting community health workers and the community itself in the identification process and establishing appropriate referral systems and treatment support. This plan will build on lessons learned through the first four packages and align with other health system strengthening and service delivery interventions in the country.

Medical Informatics; Establish disease registry

This work package aims to establish selected NCD disease registries and data-capturing tools for clinical care and epidemiological surveillance (from established community screening programs) to collect robust epidemiological data that will depict disease burden, patterns, and distribution. The data from registries will also be used to track disease trends, find people who are at high risk, evaluate care treatment and quality metrics, and evaluate treatment protocols and interventions. The project will try to fit in with the existing reporting framework as much as possible and will work with the relevant Ministry (HMIS) department.

Monitoring and evaluation

Monitoring and evaluation (M&E) is an important part of putting the project into action. A set of indicators will be made to help evaluate how well the project is being put into action and whether or not it is cost-effective enough to be scaled up. There will be standardized tools and indicators used, such as the universal health coverage indicators for type 1 diabetes, SCD, and RHD. This work package will be implemented as a part of the medical informatics package.

OUTCOME MEASURES

We will use the Proctor Implementation framework and IOM standards of care (Proctor 2011) to understand the implementation outcomes including acceptability, adoption, appropriateness, feasibility, fidelity, penetration, and sustainability. In addition to the implementation outcomes, we will also assess changes in services and client outcomes as further outlined in table 6 below.

Primary outcome measures

At the health facilities:

service capacity (measured on the day of visit using a structured questionnaire)

- Availability of functional equipment
- Availability of written guidelines and protocols
- Availability of trained staff
- Proportion of health workers present at the clinics on the day of assessment
- Proportion of tracer drugs and consumables present and non expired on the day of assessment
- Capacity to obtain basic tracer laboratory results within one day

Service utilization (data collected from facility registers and reports)

- Number of outpatient visits for a number of tracer conditions
- Number of clients screened for tracer conditions

Service Quality (Clinic and patient record audit, exit interviews)

Proportion of patients treated according to protocol

Patients' satisfaction with services provided

Patient outcome of treatment; proportion of patients reaching treatment goals for the tracer diseases and conditions
Proportion of patients with conditions being hospitalized, length of hospital stay
Case fatality rates

At Community level: (cross sectional survey)

- Proportion of community members knowledgeable about NCD risk factors and are implementing a healthy lifestyle
- Prevalence of risk factors among the community members

Routine Monitoring and Evaluation

The M&E section of the PEN-Plus implementation involves quarterly review and reflection on collected information to inform formative, strategic, evidence-based improvements to the implementation efforts. As site M&E teams send in indicator data, the data will be checked for errors, missing information, and outliers. The team will talk to the teams at each site to find answers to questions about data quality and completeness. Also, the team will meet with each site every three months to talk about trends and find and fix problems with data collection. These calls will also give implementers a chance to talk about their experiences on the ground, their thoughts, and possible ways to improve quality.

Service utilization indicators will include (by condition and health facility):

- Total patients ever enrolled
- Total patients currently enrolled and active in care
- Total patients <18 years old
- Total patients who are female
- Active patients who had a visit in last quarter
- New patients enrolled in last quarter
- Patients who have defaulted in last quarter
- Patient deaths reported in last quarter

Training Indicators

Quarterly data will be collected on the number of providers (by cadre) who are trained and able to offer PEN-Plus services. We will have a core set of quarterly indicators and will add other indicators as the systems mature. We will gather information, including:

- Midlevel providers fully trained for PEN-Plus
- Midlevel providers fully trained and currently working in PEN Plus clinic(s)
- Midlevel providers currently enrolled in classroom or didactic training
- Midlevel providers currently enrolled in e-learning course
- Midlevel providers currently enrolled in in-person didactic course
- Mentorship visit days this quarter to the PEN-Plus clinic(s) by specialists (total)
- Trained providers working in PEN-Plus clinics 6 months post training

Retrospective Record Reviews

We will document the site-specific patient cohorts using retrospective record reviews or electronic record analysis (depending on site capability). Anticipating that a majority of sites will begin with paper-based records, a retrospective review will be performed once during implementation and once at the end of the three-year period. In order to do retrospective record reviews, routine patient records will be looked at by hand and information on patient outcomes for

key conditions will be collected.
Data from the patient cohort will include:

- Number of patients being treated
- Costs to the patient
- Availability of PEN-Plus to the poorest and sickest patients
- Patient progress
- Patient-centeredness
- Clinical characteristics and outcomes

Formative discussion/in-depth interviews:
Formative discussions will be conducted with NCD leaders at the hospitals, and will have no formal interview guide, but will just be a time for leaders to air their concerns and talk about what they think is going well, or where they need more support

Cost Analysis
We will review payroll, budgets, and other financial documentation available at each facility. Based on this information, we will estimate the total annualized cost of services for all patients by adding up the estimated costs for several cost categories, such as personnel, equipment, infrastructure, and consumables. Coupled with outpatient and inpatient service volumes, we will then generate an average cost per patient per visit for outpatient services and an average cost per patient per day for inpatient services.

Table 5: Summary of Outcome measures

Question	Implementation outcome(s)	Service outcome(s)	Client outcome(s)	Evaluation component
Clinical Implementation Outcomes				
Are PEN-Plus clinics enrolling sufficient numbers of patients largely with severe chronic noncommunicable diseases?	Fidelity Penetration	Effectiveness Equity		Retrospective record reviews
Is the care being provided at these new clinics acceptable and comparable in quality to tertiary chronic care facilities?		Safety Effectiveness	Function Symptomatology Satisfaction	Formative assessments, M&E, published reports
Are patients enrolled in these clinics being protected from further impoverishment?		Equity	*Access* *Financial burden*	Retrospective record reviews

What is the cost of providing these services?	Costs			Cost analysis
Are patients' voices being heard?		Patient centeredness		Retrospective record reviews
Training Outcomes				
What is the readiness of the supported facilities to become training centers for PEN-Plus?	Adoption Appropriateness Feasibility Fidelity Penetration Sustainability			Baseline, M&E, Training, formative assessments, midline, endline, policy interviews
How many midlevel providers have been prepared to become PEN-Plus master trainers including a sufficient period of clinical practice precepted by internal medicine specialists, pediatricians, cardiologists, and endocrinologists	Adoption Fidelity Penetration			M&E, Training
What is the six-month retention of these midlevel providers?	Sustainability			Training
Policy Outcomes				
Has the Ministry of Health, together with its partners, developed the policies and strategies needed to enable implementation of PEN-Plus services at a national scale?	Acceptability Adoption Penetration Sustainability			Policy interviews, Monitoring of MoH activities

Study implementation/ Management Framework

Study implementation Team

The National Institute for Medical Research (NIMR) will oversee and coordinate the project, which will be done with the help of a group of domestic partners. NIMR will work with Muhimbili National Hospital, Jakaya Kikwete Cardiac Institute (JKCI), Muhimbili University of Health and Allied Sciences (MUHAS), Dodoma (Benjamin Mkapa) Referral Hospital, Tanzania Diabetes Association (TDA), Tanzania Sickle Cell Foundation, and CUAMM to use the strengths of the country's clinical centers of excellence to develop clinical standards, provide effective mentoring, and provide high-quality clinical care. Lastly, TANCDA will be a key partner in the planning and implementation processes to make sure that the approach is patient-centered and takes into account the wants and needs of people with NCDs.

Project team

Table 6: List of project investigators in Tanzania and technical working group (TWG) members

TWC	Name	Organization	Position
Research & Data Management	Dr Mary Mayige	NIMR	Member
	Prof Julie Makani	MUHAS	Member
	Dr Amani Mori	MUHAS	Member
	Mr.Gibson Kagaruki	NIMR	Chairperson
	Dr Peter Kisenge	JKCI	Member
	Dr Willfredius Rutahoile	Benjamin Mkapa Hospital	Member
	Dr Bruno Sunguya	MUHAS	Member
	Dr Reuben Mutagaywa	MUHAS/JKCI	Member
Training and clinical guidelines	Dr Pilly Chillo	MUHAS	Member
	Dr Edna Majaliwa	MNH	Member
	Dr Aidan Banduka	MNH	Member
	Prof Kajiru Kilonzo	KCMC	Chairperson,
	Dr Sarah Urasa	KCMC	Member
Policy & Advocacy	Dr Deogratias Soka	TANSCDF	Member
	Dr Michelle D'Alessandro	DOCTORS WITH AFRICA CUAMM	Member
	Dr Esther Mtumbuka	CHAI	Chairperson
	Prof Kaushik Ramaiya	TANCDA/ TDA	Member
	Dr David Raymond Mwenesano	NHIF	Member

Study Management Structure

Steering Committee.

The steering committee will be made up of the heads of the participating institutions or their representatives from MUHAS, NIMR, CHAI, MNH, JKCI, KCMC, Doctors with Africa, CUAMM, Dodoma Referral Hospital, Doctors with Africa, KCMC, JKCI, and KCMC, as well as the project's lead researcher. The role of the steering committee is to ensure the successful implementation of the project. The committee will meet biannually and receive reports quarterly.

Data Management Committee.

The PI and co-investigators will be on the data management committee. The committee will meet every three months to look at the quality of the data, oversee data analysis plans, and approve publications and technical reports for the project.

Technical Working Groups (TWGs).

The project will have working groups (WG) that will be led by the WG chair, who will report to the project PI. The WG chairs will oversee the implementation of the project's activities. The WG will also be in charge of training trainers and ensuring that care and treatment are based on protocols at the selected facilities.

Technical Working Groups

Training Materials Development and Clinical Services

This group will make sure that training materials and treatment plans for preventing and treating the three chosen conditions at different levels of care are made and that they are followed. It will also be responsible for training trainers and making sure that care and treatment at the chosen facilities are based on protocols. The WG will address supply chain management, patient recruitment, and follow-up.

Research Data Management and M&E

As part of the research and data management, the M&E framework for the proposed project will be made and put into place. The team will be in charge of coordinating and running the need assessment and subsequent surveys that will be used in the project. They will also be in charge of coordinating the development of study protocols, data collection plans, and platforms; creating and using the M&E framework; and coordinating the overall production of reports and publications.

System strengthening, Community Engagement, Policy and Advocacy

This WG will be in charge of coordinating all community programs. They will create the community engagement and advocacy strategy and implementation plans, including high-level advocacy with policymakers, and coordinate partnerships with stakeholders. This work plan calls for working with the appropriate health management teams at the LGA level and addressing bottlenecks in the supply of identified NCD-related commodities for the implementation of the project. Their role will also be to engage in advocacy and peer support groups. The working group will also oversee the implementation of the other community-based interventions. This working group will make a costed plan for a possible national scale-up of this model based on what they learned and what tools they made during the pilot stage.

Project Site Committees.

At each level of implementation, a local committee will be made up of the council medical officer, the medical office in charge of the facility, community development officers, and representatives from community leaders or groups that help patients.

Project Activities and Budget and Expected Outcomes

This section describes project activities, required inputs, and an evaluation framework. These are summarized in tables 8, 9, and 10 below.

Table 7 **Project Logical Framework**

Program Goal: To generate evidence and costed operational plans to scale up of the WHO Package of Essential Non Communicable Interventions (WHO PEN) for Common NCDs along with PEN PLUS interventions for selected severe NCDs type 1 diabetes, rheumatic heart diseases and sickle cell disease at district (first level) hospitals in Tanzania			
Strategic objective	**Outputs**	**Activities**	**Inputs**
To establish a PEN-Plus training center to support PEN-Plus replication in other project areas and subsequent roll out of Integrated care services for NCDs;	Stakeholders are fully engaged at all stages of the project	National level meeting	Travel costs, conference package, personnel reimbursement
		Zonal level meetings	Travel costs, conference package, personnel reimbursement
		District level meetings/ Community level stakeholder engagement meetings	Travel costs, conference package, personnel reimbursement
	Project is well coordinated and implemented accordingly	Steering committee and technical working group meetings	Travel costs, conference package, personnel reimbursement
		Project staff - National coordinator - Project admin - Site coordinator	Personnel emoluments
	Facility baseline assessment survey done and data used for planning	Need assessment survey	Consultancy to conduct survey
		Feedback meeting / planning	Travel costs, conference package, personnel reimbursement
	PEN PLUS facilities are well equipped and functional	Purchase and fitting of equipments	Funds for procurement
		Minor Facility Renovation	Funds for renovation
		Establish supply chain	Travel costs, conference package, personnel reimbursement
		Establish framework for patient follow up and M&E including IT infrastructure and disease registries	Consultancy to build and maintain data system IT personnel emoluments IT equipment and software Registry maintenance costs

	Training resources are available, and training is conducted as planned.	Develop training materials	Travel costs, conference package, personnel reimbursement
		Clinical guidelines development	Travel costs, conference package, personnel reimbursement
		Conduct training	Travel costs, conference package, personnel reimbursement
		Establish Mentorship programs for onsite training	Travel costs, personnel reimbursement
To pilot PEN-Plus services in 3 first-referral level hospital facilities	Sufficient Patients are recruited and followed up at the PEN PLUS Clinics	Ensure patient enrolment , treatment and followup care in the established PEN PLUS clinic	Medication , diagnostics and referral costs
	Personnel (clinical and supportive staff) are available as needed	Personnel management and emolument	Personnel emoluments
To generate demand for the introduced PEN-Plus services in population around the identified 3 first-referral level hospital facilities	Outreach to local health facilities including strenthening referrals are conducted accordingly	Supervision/ mentorship visits to lower level facilities	Travel costs, personnel reimbursement
	Community assessment is done and data utilized to inform health education interventions	Baseline Knowledge Attitude and Practice (KAP) survey	Consultancy to conduct survey
		Post intervention KAP survey	Consultancy to conduct survey
	Health education materials developed and piloted	Conduct workshops to develop and adapt health education materials Printing and distribution	Travel costs, conference package, personnel reimbursement, printing and distribution costs
	Community awareness campaigns are	Community awareness activities	Travel costs, personnel reimbursements including CHWs, Media coverage costs

	conducted acocrdinly		

Budget Summary and Justification

Below is a summary of the amount of funds requested to implement the PEN PLUS project in 2 sites over 3 years.

Table 8: Budget

Outcome	**Activity**	**Inputs**	**Budget**		
			Y1	**Y2**	**Y3**
Stakeholders engagement	National level meeting	Travel costs, conference package, personnel reimbursement	$ 15,446	$ 15,000	N/A
	District level meetings/ Communty level stakeholder engament meetings	Travel costs, conference package, personnel reimbursement	$ 1,360	$ N/A	N/A
Subtotal			**$16,806**	**$ 15,000**	**N/A**

Project coordination	Steering committee and technical working group meetings	Travel costs, conference package, personnel reimbursement	$ 15,600	$ 15,300	$ 17,100
	Project staff	Personnel emoluments	$ 120,900	$ 154,500	$ 154,500
Subtotal			**$136,500**	**$169,800**	**$169,800**
Facility baseline assessment survey	Need assessment survey	Costs to conduct survey	$ 10,000	N/A	N/A
	Feedback meeting / planning	Travel costs, conference package, personnel reimbursement	$ 2,640	N/A	N/A
Subtotal			$ 1**2,640**	**$00**	**$00**
Equip facilty for PEN PLUS	Purchase and fitting of equipments	Funds for procurement	$ 26,000	$ 28,000	N/A
	Minor Facility Renovation	Funds for renovation	$ 10,000	$ 7,000	$ 10,000

	Establish supply chain	Travel costs, conference package, personnel reimbursement	$ 4,010	$7,984	$ 7,572
	Establish framework for patient follow up and M&E including IT infrastructure	Costs to build and maintain data system	$ 10,000	N/A	N/A
		IT personnel emoluments			
		IT equipment and software			
Subtotal			**$50,010**	**$ 42,984**	**$17,572**
Training	Develop training materials	Travel costs, conference package, personnel reimbursement	$ 7,925	N/A	N/A
	Clinical guideliness development	Travel costs, conference package, personnel reimbursement	$ 9,525	x	X

	Conduct training	Travel costs, conference package, personnel reimbursement	$ 19,900	$ 3,980	N/A
	Establish Mentroship programs for onsite training	Travel costs, personnel reimbursement	$ 26,064	$ 11,880	$,5,700
Subtotal			**$63,414**	**$15,860**	**$ 5,700**
Patient recruitment and follow up	Ensure patient enrolment , treatment and followup care in the established PEN PLUS clinic and screening	Medication , diagnostics and referral costs	$ 34,800	$ 87,000	$ 127,020
		RHD patient Surgery costs		$33,000	$63,000
Personnel (clinical and supportive staff)	Personnel management and emolument	Personnel emoluments	Item costed above	Item costed above	Item costed above

Outreach to local health facilities including strengthening referrals	Supervision/ mentorship visits to lower level facilities	Travel costs, personnel reimbursement	$ 2,160	$ 4,320	$ 4,320
Subtotal			**$36,960**	**$124,320**	**$ 194,340**
Community assessment	Baseline survey	Costs to conduct survey	$ 5,000	N/A	N/A
	Post intervention survey	Costs to conduct survey	$ 5,000	N/A	N/A
Develop and pilot Health Education materials	Conduct workshops to develop and adapt health education materials	Travel costs, conference package, personnel reimbursement, printing and distribution costs	$ 7,280	N/A	N/A
	Printing and distribution				
Community awareness campaigns	Community awareness activities and screening	Travel costs, personnel reimbursements including CHWs,	$ 7,420	$ 21,066	$ 16,840

		Media coverage costs			
Subtotal			**$24,700**	**$21,066**	**$16,840**
Conference and Regional and international meetings attendance	Publications	Publication fees	N/A	N/A	$ 2,000
	Conference Registration	Conference registration fees	N/A	N/A	$ 2,000
	International Travel	Travel costs	N/A	N/A	$ 10,000
Final Reporting and drafting of National PEN PLUS implementation plan	Conduct workshops	Travel costs, conference package, personnel reimbursement	N/A	N/A	$ 5,378
Final Report dissemination	Report Printing and Distribution	Printing and distribution costs	N/A	N/A	N/A
	National Results Dissemination Workshop and report launch	Travel costs, conference package, personnel reimbursement	N/A	N/A	$ 15,350
Subtotal			$ **00**	**$00**	**$ 34,728**

Grand Total			**$ 341,430**	**$ 389,030**	**$438,220**

Table 9: Budget Justification

	Budget Line	Amount	Justification
1.	Salaries	408,300$	A total of 408,300$ and 21,600$ is requested to support salaries and fringe benefits for project staff for both the national level coordination and for implementation at sites including consultant reimbursements to support lower level facilities. Remuneration is based on their salaries and level of efforts as outlined above
	Fringe benefits	21,600$	
2.	Equipment & Supplies	60,478$	A total of 60,478$ is requested for equipment and supplies. These are costs related to purchase and installation of equipments itemized in table 4 as needed in the 2 initial sites
3.	Other direct costs (coordination, mentorship , and advocacy)	235,385$	A total of 235,385$ is requested for per diem and transport cost related to project national level meeting ,mentorship onsite training ,dissemination workshop for PEN PLUS final report to stakeholders(national level) and community awareness
4	Other direct costs (patient follow up and referral , supplies and costs of surgical procedures for severe RHD patients who require surgery	442,877$	A total of 442,877$ is requested to cover all cost related to operation surgery,minor repair of clinics and procurement medical supplies and fittings of equipments and follow up of patients
5	Other indirect costs	116,864$	A total of 116,864$ is requested as 10% institutional cost
	Total Project Costs	1,285,504$	A total of 1,285,504$ is requested to implement PEN PLUS Project in 2 sites over 3 years

Table 10: PEN-Plus Project Initiation and Implementation: Preliminary Deliverables Framework

Core Activities	Targets/Deliverables	20
PEN-Plus Clinic Implementation & Service Delivery		
Design and plan PEN-Plus clinical care model adapted to the local context	Model designed	Y1
Implement PEN-Plus clinics as demonstration sites	# of PEN-Plus demonstration sites	Y1, Y2
Enroll patients in PEN-Plus clinics after three years	# of patients enrolled	Y1, Y2, Y3
Community awareness campaigns	# of activities conducted	Y1, Y2, Y3
PEN-Plus Human Resources, Training & Mentorship		
Train mid-level providers to lead PEN-Plus clinical care	# of providers trained	Y1, Y2
Implement support for PEN clinics at level by PEN-Plus clinics	# of PEN clinics supported	Y2, Y3
Establish national training and mentorship sites	# of training and mentorship sites	Y1, Y2, Y3
Incorporate sub-specialists (e.g., cardiologists, endocrinologists, etc.) into the training and mentorship of PEN-Plus providers	# of sub-specialists	Y1, Y2, Y3
PEN-Plus Information Systems & Supplies		
Design and implement routine clinical data collection forms	# of demonstration sites with data collection	Y1, Y2
Develop policies, mechanisms, and structures for procurement, distribution, and surveillance of PEN-Plus commodities; this includes routine analysis of surveillance data to support decision-making on procurement and distribution.	Establish targets	Y 1
Design and implement a PEN-Plus monitoring and evaluation framework	Framework established	Y1, Y2, Y3
PEN-Plus Policy Development/STAKEHOLDER ENGAGEMENT		
Incorporate PEN-Plus clinical protocols into national guidelines	# of clinical protocols incorporated	Y 1
Establish National PEN-Plus working group that regularly meets to lead implementation, policy, and financial aspects of the program	Working group established	Y1, Y2, Y3
Establish a draft national operational plan that describes national scale-up of PEN-Plus	Operational plan developed	Y 3
Project Coordination Meetings		
-Steering committee/ TWG / Project site meetings	# of meetings conducted	Y1, Y2, Y3



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